Narratives, Noise Traders, and Asset Prices:

A General Equilibrium Approach

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Outline

1. Introduction

2. The Model

3. Analytical Results

This Paper

- Conflicting views in literature on how noise traders affect asset prices
  1. **No lasting effect** – arbitrage undoes their trades; anomalies don't survive empirical tests
     - **Friedman (1953); Fama (1998)**
  2. **Temporary effect on prices** – prices are pushed but reverts; no persistent effect on expected returns
     - **Barber and Odean (2008); Da, Engelberg and Gao (2011)**
  3. **Effect on both prices and returns** – especially when arbitrage is limited
     - **De Long et al. (1990); Baker and Wurgler (2006)**
- **This paper**: a *general-equilibrium* model in which *noise traders* act on *narratives*
  - *Narratives*: stories about the economy that need not reflect fundamentals
  - *Noise Traders* treat narratives as fundamental; rational traders know they are noise
- **We show**: all three views may arise in the same economy, depending on the ***scope*** of the narratives in play

What are Narratives?

- Five features of narratives (Shiller 2017, 2019) documented in literature:
  1. **They arise suddenly, even without new information** (fundamental news)
  2. **They spread through contagion and fade as interest wanes**, often modeled as epidemics
  3. **They recur:** dormant stories return when interest is revived
  4. **They move demand and prices** by concentrating attention on certain stocks
  5. **Their scope varies**: some concern a single firm or a few, others entire sectors or the economy
- We model each narrative Visual as:
  1. Arriving at random (**Poisson**), independently of news
  2. **Poisson jump** in intensity when it emerges, then fades exponentially with time
  3. Recurring **random flare-ups**; more likely soon after a recent flare-up
  4. **Shifts noise traders' expected returns** on the stocks it concerns
  5. Each narrative concerns a **different set of firms (randomly drawn)**

What we Do

1. **A Cross Section of Stocks with a Factor Structure**
   - J stocks with different exposures to two Lucas trees, idiosyncratic shocks on stocks that are persistent, but mean-reverting
2. **Two investor types, in perpetual youth**
   - Noise traders act on narratives, rational traders don't
   - New noise traders keep arriving, and keep making mistakes
3. **Narratives of different scopes in circulation**
   - Stories about a firm, a theme, a style or the market – drawn from a distribution of scope
4. **Solved in General Equilibrium**
   - Calibrated to the market Sharpe ratio, demand elasticities, and idiosyncratic volatility

What we Find

1. **The mix of narratives in circulation decides what kind of distortion a market has**
   - Each narrative's scope – which firms, and how many, it affects – is drawn from a distribution
     - Statistics of that distribution set how distortion splits into firm-specific / systematic
     - Only a few of them matter
   - Markets with more "narrow" stories show price mistakes that correct, and look efficient in returns
   - Markets with "broader" stories show shifts in expected returns that look like risk
2. **The evidence budget – noise traders can leave only as much evidence as they lose**
   - More "intense" mistakes drain noise traders' wealth faster, capping what narratives reveal in returns
   - (With no frictions) Noise traders persist but stay small, return-based anomalies are weak and fragile
3. **The data point to limited arbitrage**
   - With frictions – **tracking-error limits, high risk-aversion for rational traders** – calibration targets are better matched
   - There, narratives have larger effects on prices, and aggregate demand is inelastic, as in the data

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Production

- **2 Lucas Trees** producing cash-flows D_k

$$\frac{\mathrm{d}D_k(t)}{D_k(t)} = \mu_{Dk}(\zeta)\,\mathrm{d}t + \sigma_{Dk}\,\mathrm{d}Z_{fk}(t), \qquad k = 1, 2$$

- **Mean-Reverting Shares.** $\mu_{Dk}(\zeta) = \bar{\mu} \mp \frac{\kappa_D}{2}(\zeta - \bar{\zeta})$; growth rates depending on $\zeta \equiv \ln(D_1/D_2)$
  - No tree dominates: whichever is smaller grows faster.
- J **Firms**, each owning a mix of the two trees, each representing a distinct segment in the economy:

Tree 1	Tree 2
New Ventures	Established Operations
Intangible Capital	Physical Capital

  - δ_j **(the share of firm j's value coming from tree 1)** places each firm on a spectrum between the two trees

Asset Menu

- Investors trade $J + 1$ assets
  1. A riskless bond in zero net supply, paying the rate $r(X, t)$ **(set in equilibrium)**
  2. J stocks, one per firm, each in unit supply
- For each stock $j \in \{1, \ldots, J\}$
  1. **Payout** (Π_j)**:** a share $\Upsilon_j \, s_{kj}$ of each tree's cash flows D_{kj}
     - $s_{kj} \in [0, 1]$ are the baseline shares, constant
     - Υ_j is an idiosyncratic multiplier, mean-reverting and driven by idiosyncratic Brownian $Z_{\iota j}$
  2. **Returns** (R_j)**:**

$$\mathrm{d}R_j \equiv \frac{\mathrm{d}P_j + \Pi_j \mathrm{d}t}{P_j} = \mu_{Rj}(X, t)\,\mathrm{d}t + \boldsymbol{\sigma}_{Rj}^{\top}(X, t) \underbrace{\mathrm{d}\mathbf{Z}_f}_{\text{Systematic}} + \sigma_{\iota j} \underbrace{\mathrm{d}Z_{\iota j}}_{\text{Idiosyncratic}} + \underbrace{\mathrm{d}\Xi_j}_{\text{Narratives}}$$

Demographics

1. **Two Agent Types.** A fixed fraction $\alpha \in (0, 1)$ of Noise Traders, and $1 - \alpha$ Rational Traders
2. **Perpetual Youth.** **(Yaari 1965, Blanchard 1985)**
   - Continuous-time OLG with unit mass
   - Poisson exit rate χ with matched entry flow; fixed entry composition $(\alpha, 1 - \alpha)$, preserving demographic fractions
   - No bequest, actuarially fair annuities: active agents earn $r + \chi$, the problem stays scale invariant.
3. **Entry Endowment** $W_e = \tau Y / \chi$**.** Funded by a flat tax τ on output

Narratives

- **Narratives** distort what noise traders expect firms to earn

$$\mu_{Rj}^{N} = \mu_{R,j} + \sum_{\ell} b_{\ell,j}\,\xi_{\ell}$$

  - ξ_{ℓ} measures narrative ℓ 's intensity, and $b_{\ell j}$ how strongly it applies to firm j.

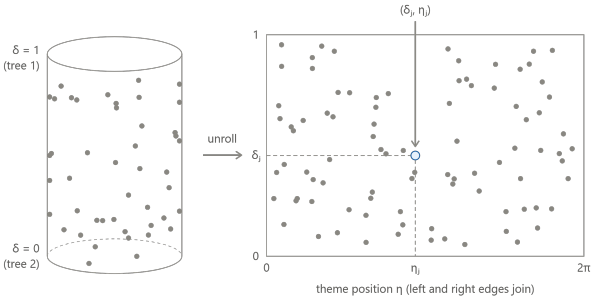
- **Dynamics of ξ_{ℓ}.**

$$\mathrm{d}\xi_{\ell}(t) = -\frac{\xi_{\ell}(t)\,\mathrm{d}t}{\varepsilon_{\ell}} + \psi_{\ell}(t)\,\mathrm{d}N_{\ell}(t)$$

  - ξ_{ℓ} spawns randomly at arrival time $\mathcal{T}_{\ell 1}$, following a Poisson process with rate χ_{ξ}
  - The counting process $N_{\ell}(t)$ records repeated "random flare-ups": $\mathbb{P}(\mathrm{d}N_{\ell}(t) = 1) = \nu_{\ell}(t)\,\mathrm{d}t$
  - The rate of flare-ups: $\nu_{\ell}(t)$ = (Popular Interest) + (Holdings). Popular interest decays over time.

Map of Stories

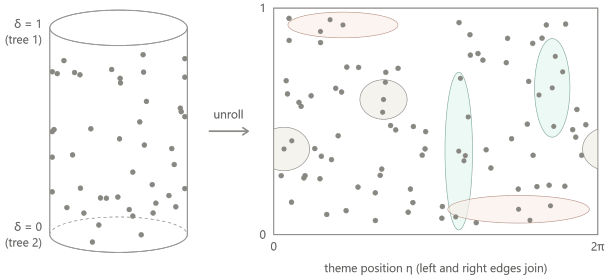
- Each firm is placed on a map of stories (δ_j, η_j). Examples



- δ_j is fundamental exposure; η_j is the firm's place on the "thematic circle", independent of fundamentals.

Map of Stories, with Narratives

- For each ℓ, random draws: *(i)* **position** $(\delta_j^\ell, \eta_j^\ell)$, and *(ii)* **reach** along the two dimensions $(\varrho_\ell^\delta, \varrho_\ell^\eta)$



- Together, position and reach determine the narratives' ***scope***. Examples

Anatomy of Narratives

The scope determines the loading vector $\mathbf{b}_\ell = (b_{\ell 1}, \ldots, b_{\ell 2})^\top$

- $b_{\ell j}$ is largest for firms close to center $(\delta_j^\ell, \eta_j^\ell)$, fading with distance on the map, and 0 for firms outside the scope

It characterizes the narrative via two channels:

1. **Channel 1 (Main) – Alignment θ_ℓ:** how much of it lies along the systematic risk where

$$\mathbf{b}_\ell \propto \cos\theta_\ell \, \mathbf{e}_\ell^\iota + \sin\theta_\ell \, \mathbf{e}_\ell^f, \qquad \theta_\ell \in [0, \pi/2]$$

   $\mathbf{e}_\ell^\iota$, $\mathbf{e}_\ell^f$ are directions of $\mathbf{b}_\ell$'s projection onto the systematic subspace, and the orthogonal complement

2. **Channel 2 – Comovement and Spillovers**
   - Comomvement: firms moving together beyond what fundamentals imply
   - Spillovers: firms outside its scope can be affected too

Optimization

- Wealth accumulates objectively (common for all investors)

$$\frac{\mathrm{d}W}{W} = \left[\boldsymbol{\pi}^\top \boldsymbol{\mu}_e + (r + \chi) - c\right] \mathrm{d}t + \boldsymbol{\pi}^\top \left(\Sigma_R \mathrm{d}\mathbf{Z}_f + \Sigma_\iota \mathrm{d}\mathbf{Z}_\iota + \mathrm{d}\boldsymbol{\Xi}\right)$$

  - $\boldsymbol{\pi}$: portfolio weights; $c \equiv C/W$: consumption/wealth; $\boldsymbol{\mu}_e \equiv \boldsymbol{\mu}_R - r$: excess expected returns

- The budget constraint used for optimizing:
  1. Rational traders use objective dynamics
  2. Noise traders use their own perceived returns, distorted by narratives ($\boldsymbol{\mu}_e \to \boldsymbol{\mu}_e^{\mathcal{N}}$)

- Type specific **Optimal Policies,** (derived from ▸ type specific HJB Equations) for $i \in \{\mathcal{N}, \mathcal{R}\}$,

$$\boxed{c^{i*} = v^i(X,t)^{-1/\gamma}, \qquad \boldsymbol{\pi}^{i*} = \frac{1}{\gamma}\Omega^{-1}\boldsymbol{\mu}_e^i + \frac{1}{\gamma}\Omega^{-1}\Sigma_{RX}\nabla \ln v^i(X,t)}$$

  - Ω is the return covariance matrix (systematic + idiosyncratic risk)

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Some Notation

1. Type specific wealth $W^{\mathcal{N}}, W^{\mathcal{R}}$ and the noise trader wealth share $\omega \equiv W^{\mathcal{N}}/(W^{\mathcal{N}} + W^{\mathcal{R}})$
2. Wealth weighted averages: $y^{M} = \omega\, y^{\mathcal{N}} + (1-\omega)\, y^{\mathcal{R}}$
   - *(e.g.)* $\boldsymbol{\pi}^{M} = \omega\, \boldsymbol{\pi}^{\mathcal{N}*} + (1-\omega)\, \boldsymbol{\pi}^{\mathcal{R}*}, \qquad c^{M} = \omega\, c^{\mathcal{N}*} + (1-\omega)\, c^{\mathcal{R}*}$
3. Disagreement: $\Delta y = y^{\mathcal{N}} - y^{\mathcal{R}}$
   - *(e.g.)* $\Delta\boldsymbol{\pi} = \boldsymbol{\pi}^{\mathcal{N}*} - \boldsymbol{\pi}^{\mathcal{R}*}, \qquad \Delta c = c^{\mathcal{N}*} - c^{\mathcal{R}*}$
4. Because (1) the risk-free asset is in zero net supply, and (2) the risky asset markets clear:

$$\underbrace{\bar{\pi}_j}_{\text{Aggregate Demand}} = \underbrace{\frac{P_j}{P}}_{\text{Market Portfolio}}, \qquad j = 1, \ldots, J$$

   - $P = \sum_j P_j$ is the total market value.

Workhorse Result 1: Trading Capacity and Capitalization

1. **Relative Trading Capacity** ϖ of noise traders

$$\varpi = \frac{\omega/\tilde{\gamma}^{N}}{1/\tilde{\gamma}^{M}} = \omega\frac{\tilde{\gamma}^{M}}{\tilde{\gamma}^{N}}$$

   - *Effective* risk aversion $\tilde{\gamma}^{i}$: how reluctant type i is to act on expected returns ($\boldsymbol{\pi}^{i} = \frac{1}{\tilde{\gamma}^{i}}\Omega^{-1}\boldsymbol{\mu}_{e}^{i}$)
     Capturing, in reduced form, market frictions $e.g.$ risk aversion, tracking-error limits, short-sale costs
   - As $\varpi \to 0$, rational traders have higher capacity to arbitrage
   - With homogeneous risk aversion γ and no additional frictions, $\tilde{\gamma}^{N} = \tilde{\gamma}^{M} = \gamma$, so $\varpi = \omega$

2. **Capitalization.** market clearing passes a share ϖ of the belief distortion into expected returns

$$\boldsymbol{\mu}_{e} = \boldsymbol{\mu}_{e}^{*} - \varpi\,\Delta\boldsymbol{\mu}_{e}, \qquad \Delta\boldsymbol{\mu}_{e} = \sum_{\ell}\mathbf{b}_{\ell}\,\xi_{\ell}$$

   - $\boldsymbol{\mu}_{e}^{*}$ is expected excess returns without narratives

Workhorse Result 2: The Narrative Distortion $\mathcal{G}$

- Write $\mathcal{G} \equiv \mathbb{E}\left[\|\Delta\boldsymbol{\mu}_e\|^2_{\Omega^{-1}}\right]$ for the distortion caused by narratives, measured in Sharpe-ratio units:
  - How large an *opportunity* noise traders *perceive* beyond what the market offers, averaged over time

$$\mathcal{G} \equiv \mathbb{E}\left[\Delta\boldsymbol{\mu}_e^\top \Omega^{-1} \Delta\boldsymbol{\mu}_e\right] = \mathbb{E}\left[\sum_\ell \sum_m \xi_\ell \xi_m \mathbf{b}_\ell^\top \Omega^{-1} \mathbf{b}_m\right]$$

$$= \underbrace{\mathbb{E}\left[\left\|\sum_\ell \xi_\ell \sin\theta_\ell \, \mathbf{e}_\ell^f\right\|^2_{\Omega^{-1}}\right]}_{\mathcal{G}^f\text{: systematic distortion}} + \underbrace{\mathbb{E}\left[\left\|\sum_\ell \xi_\ell \cos\theta_\ell \, \mathbf{e}_\ell^\iota\right\|^2_{\Omega^{-1}}\right]}_{\mathcal{G}^\iota\text{: firm-specific distortion}}$$

- $\mathcal{G}^f$ and $\mathcal{G}^\iota$ are two quantities through which narratives enter the results that follow
  - $\mathcal{G}^f$ is the opportunity, perceived by noise traders, in betting more/less on the market as a whole
  - $\mathcal{G}^\iota$ is the opportunity, perceived by noise traders, in picking particular firms over others

Workhorse Result1: From Scope to Distortion

Suppose

$$\begin{pmatrix} \ln \varrho_\ell^\delta \\ \ln \varrho_\ell^\eta \end{pmatrix} \sim \mathcal{N}\left(\begin{pmatrix} \mu_\delta \\ \mu_\eta \end{pmatrix}, \begin{pmatrix} \sigma_\delta^2 & \rho_{\delta\eta}\sigma_\delta\sigma_\eta \\ \rho_{\delta\eta}\sigma_\delta\sigma_\eta & \sigma_\eta^2 \end{pmatrix} \right)$$

- The systematic share:

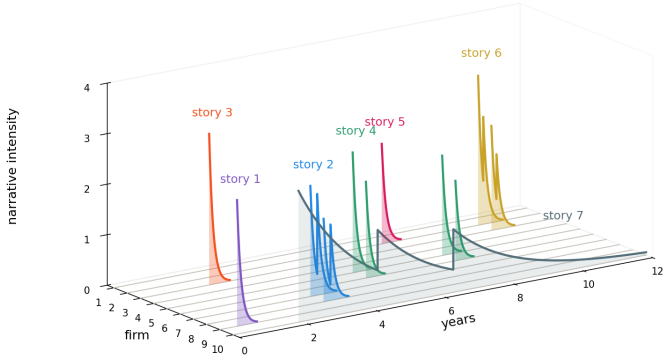
$$\ln\left(\frac{\mathcal{G}^f}{\mathcal{G}}\right) \approx (\text{const}) + \mu_\delta + \mu_\eta + \frac{1}{2}(\sigma_\delta^2 + \sigma_\eta^2 + 2\rho_{\delta\eta}\sigma_\delta\sigma_\eta)$$

- The systematic share rises with the mean reaches, their dispersions, and their correlation

Thank You

Thank You

Narratives: Visuals



return

Map of Stories: Examples

return

1. **Fundamentals.** Tree 1: New Ventures (**high** μ, **high** σ); Tree 2: Established Operations (**low** μ, **low** σ)
   - Higher δ_j: *Growth* firms; lower δ_j: *Value* firms
2. **Themes.** Consumer-facing industries

Angle η	Theme
$0 = 2\pi$	Food and beverages
$\pi/2$	Health and fitness
π	Entertainment and media
$3\pi/2$	Home and living

- $(\delta_1, \eta_1) = (0.8, 3\pi/4)$: a small fast growing firm producing fitness apps and games
- $(\delta_2, \eta_2) = (0.1, 0)$: a large value firm producing canned soup
- $(\delta_3, \eta_3) = (0.5, \pi/4)$: a mid-firm with an established sports-drink line, venturing into protein bars

Narratives: Examples

return

Consider the following narratives, centered at $(\delta_2, \eta_2) = (0.1, 0)$, **a large firm producing canned soup**

1. Vertically Long Narrative (high ϱ^δ) – **Thematic** Narratives
   - "All food firms will suffer due to Wegovy pills"
2. Horizontally Wide Narrative (high ϱ^η) – **Style** Narratives
   - "Heritage brands are cool again"

Narratives with very high ϱ^δ, $\varrho^\eta \to \infty$ apply to all firms in the market ($\mathbf{b}_\ell \propto \mathbf{1}$)

1. Market Narrative: "It feels like a recession is coming"
   - Assume frequent & small flare-ups, coverges to OU: $\mathrm{d}\xi_M = -\frac{\xi_M}{\varepsilon_M}\mathrm{d}t + \sigma_M \mathrm{d}Z_M$ **(as in DeLong et al. 1990)**
2. Sector Narrative: "Cheap money will fuel the next wave of growth companies"
   - All firms experience a boost, but at different scales: firm 1 the most, firm 2 the least

Hamilton-Jacobi-Bellman Equation

Solving Type-i Agent's Problem: $V^i(W, X, \xi, t) = \max_{\{c,\pi\}} \mathbb{E}_t^i \left[\int_t^\infty e^{-(\rho+\chi)(u-t)} \frac{(cW)^{1-\gamma}}{1-\gamma} \, \mathrm{d}u \right]$

- HJB (with $\hat{\Omega}$ the total covariance of returns and $\Sigma_{R\hat{X}}$ their covariance with the state)

$$0 = \max_{c,\pi} \left\{ \frac{(cW)^{1-\gamma}}{1-\gamma} - (\rho+\chi)V^i + [\pi^\top \mu_e^i + (r+\chi) - c] W \partial_W V^i \right.$$
$$\left. + \frac{1}{2} \pi^\top \hat{\Omega} \pi W^2 \partial_{WW} V^i + \mathcal{L}_{\hat{X}}^i V^i + W \pi^\top \Sigma_{R\hat{X}} \partial_{W\hat{X}} V^i \right\},$$

- Reduced HJB with $V^i = \left(W^{1-\gamma}/(1-\gamma)\right) v^i$

$$\frac{\mathcal{L}_{\hat{X}}^i v^i}{v^i} + c^i - (\rho+\chi) + (1-\gamma) \left[(\pi^i)^\top \mu_e^i + r + \chi - c^i - \frac{\gamma}{2} (\pi^i)^\top \hat{\Omega} \pi^i + (\pi^i)^\top \Sigma_{R,\hat{X}} \nabla \ln v^i \right] = 0$$

return